



SIMATS ENGINEERING

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Student Name: Samiksha Dhanya G

Reg. No.: 192321014

Course Code: CSA1504

Slot: A

Course Name: Cloud computing and Big data analytics

Course Faculty: DR. N Poongavanam

One-Page Individual Reflection

TITLE: Smart Energy-Consumption Analytics Using Big Data Technologies.

Individual Reflection: Samiksha Dhanya G (Reg. No: 192321014)

Core Technical Decisions & System Architecture: Developing this distributed Smart Energy-Consumption Analytics System has been a transformative hands-on engineering experience, successfully bridging the theoretical principles of cloud computing and distributed systems with practical Big Data software engineering. My primary technical focus encompassed designing the distributed storage architecture in HDFS, executing data ingestion pipelines over 2,075,259 real-world smart meter records, and authoring the core Java MapReduce engine (SmartEnergyAnalytics.java) alongside Analysis A, B, and C.

Algorithmic Engineering & Shuffle-and-Sort Optimization: A deliberate architectural choice I made was to build the core analytics using native Java MapReduce constructs rather than immediately relying on higher-level SQL abstractions like Apache Hive. This forced me to deeply engage with the internal mechanics of HDFS block partitioning (128 MB blocks), custom InputSplit record readers, and the Shuffle & Sort grouping phase. Designing the MonthlyMapper and PeakHourMapper required careful index parsing of raw CSV fields. Implementing custom in-memory Combiners before data spilled to disk significantly minimized intermediate I/O and network transfer during the reduce phase.

Challenges Encountered & Engineering Solutions: The primary technical obstacle I encountered was handling corrupted records and header anomalies across the 171.88 MB dataset. In initial cluster test runs, unhandled NumberFormatExceptions on missing values caused entire map tasks to terminate abruptly. I resolved this by implementing defensive try-catch wrappers, rigorous null-field guards, and robust boundary checks inside the mapper's map() method, enabling the distributed pipeline to process over two million records flawlessly.

Attainment of Course Outcomes & Societal Impact: This project directly facilitated the attainment of Course Outcomes CO1, CO2, and CO3. I gained thorough competency in provisioning pseudo-distributed Hadoop

clusters, managing HDFS directories via CLI commands, orchestrating batch jobs through YARN, and interpreting analytical outputs—such as identifying that the system-wide peak load occurs precisely at 20:00 (2,720.36 kWh). From an ethical and societal standpoint, aligning our findings with SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action) demonstrated how big data analytics can modernize municipal electrical grids and accelerate renewable energy transitions.

Student Signature

Guide Signature